

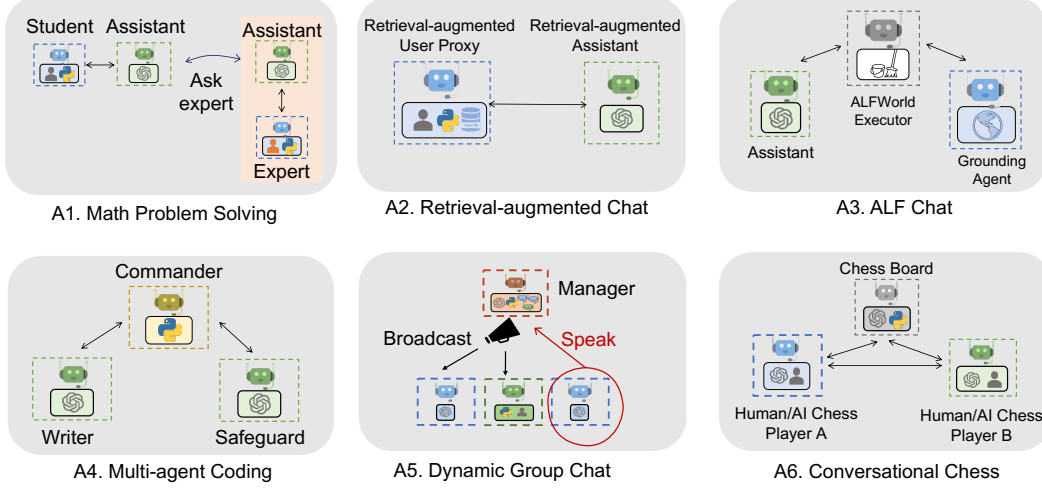


Figure 3: Six examples of diverse applications built using AutoGen. Their conversation patterns show AutoGen’s flexibility and power.

A1: Math Problem Solving

Mathematics is a foundational discipline and the promise of leveraging LLMs to assist with math problem solving opens up a new plethora of applications and avenues for exploration, including personalized AI tutoring, AI research assistance, etc. This section demonstrates how AutoGen can help develop LLM applications for math problem solving, showcasing strong performance and flexibility in supporting various problem-solving paradigms.

(Scenario 1) We are able to build a system for autonomous math problem solving by directly reusing two built-in agents from AutoGen. We evaluate our system and several alternative approaches, including open-source methods such as Multi-Agent Debate (Liang et al., 2023), LangChain ReAct (LangChain, 2023), vanilla GPT-4, and commercial products ChatGPT + Code Interpreter, and ChatGPT + Plugin (Wolfram Alpha), on the MATH (Hendrycks et al., 2021) dataset and summarize the results in Figure 4a. We perform evaluations over 120 randomly selected level-5 problems and on the entire⁵ test dataset from MATH. The results show that the built-in agents from AutoGen already yield better performance out of the box compared to the alternative approaches, even including the commercial ones. **(Scenario 2)** We also showcase a human-in-the-loop problem-solving process with the help of AutoGen. To incorporate human feedback with AutoGen, one only needs to set `human_input_mode='ALWAYS'` in the `UserProxyAgent` of the system in scenario 1. We demonstrate that this system can effectively incorporate human inputs to solve challenging problems that cannot be solved without humans. **(Scenario 3)** We further demonstrate a novel scenario where *multiple* human users can participate in the conversations during the problem-solving process. Our experiments and case studies for these scenarios show that AutoGen enables better performance or new experience compared to other solutions we experimented with. Due to the page limit, details of the evaluation, including case studies in three scenarios are in Appendix D.

A2: Retrieval-Augmented Code Generation and Question Answering

Retrieval augmentation has emerged as a practical and effective approach for mitigating the intrinsic limitations of LLMs by incorporating external documents. In this section, we employ AutoGen to build a Retrieval-Augmented Generation (RAG) system (Lewis et al., 2020; Parvez et al., 2021) named Retrieval-augmented Chat. The system consists of two agents: a Retrieval-augmented User Proxy agent and a Retrieval-augmented Assistant agent, both of which are extended from built-in agents from AutoGen. The Retrieval-augmented User Proxy includes a vector database (Chroma,

⁵We did not evaluate ChatGPT on the whole dataset since it requires substantial manual effort and is restricted by its hourly message-number limitation. Multi-agent debate and LangChain ReAct were also not evaluated since they underperformed vanilla GPT-4 on the smaller test set.

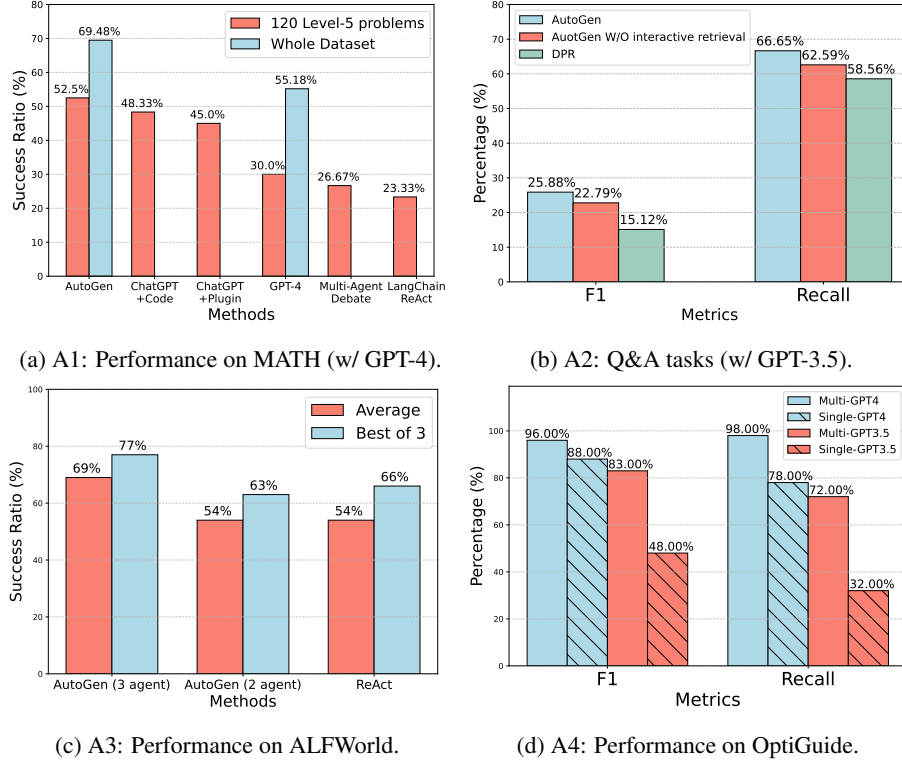


Figure 4: Performance on four applications A1-A4. (a) shows that AutoGen agents can be used out of the box to achieve the most competitive performance on math problem solving tasks; (b) shows that AutoGen can be used to realize effective retrieval augmentation and realize a novel interactive retrieval feature to boost performance on Q&A tasks; (c) shows that AutoGen can be used to introduce a three-agent system with a grounding agent to improve performance on ALFWorld; (d) shows that a multi-agent design is helpful in boosting performance in coding tasks that need safeguards.

2023) with SentenceTransformers (Reimers & Gurevych, 2019) as the context retriever. A detailed workflow description of the Retrieval-augmented Chat is provided in Appendix D.

We evaluate Retrieval-augmented Chat in both question-answering and code-generation scenarios. **(Scenario 1)** We first perform an evaluation regarding natural question answering on the Natural Questions dataset (Kwiatkowski et al., 2019) and report results in Figure 4b. In this evaluation, we compare our system with DPR (Dense Passage Retrieval) following an existing evaluation⁶ practice (Adlakha et al., 2023). Leveraging the conversational design and natural-language control, AutoGen introduces a novel *interactive retrieval* feature in this application: whenever the retrieved context does not contain the information, instead of terminating, the LLM-based assistant would reply “*Sorry, I cannot find any information about... UPDATE CONTEXT.*” which will invoke more retrieval attempts. We conduct an ablation study in which we prompt the assistant agent to say “*I don’t know*” instead of “*UPDATE CONTEXT.*” in cases where relevant information is not found, and report results in Figure 4b. The results show that the interactive retrieval mechanism indeed plays a non-trivial role in the process. We give a concrete example and results using this appealing feature in Appendix D. **(Scenario 2)** We further demonstrate how Retrieval-augmented Chat aids in generating code based on a given codebase that contains code not included in GPT-4’s training data. Evaluation and demonstration details for both scenarios are included in Appendix D.

⁶The results of DPR with GPT-3.5 shown in Figure 4b are from (Adlakha et al., 2023). We use GPT-3.5 as a shorthand for GPT-3.5-turbo.

A3: Decision Making in Text World Environments

In this subsection, we demonstrate how AutoGen can be used to develop effective applications that involve interactive or online decision making. We perform the study using the ALFWorld (Shridhar et al., 2021) benchmark, which includes a diverse collection of synthetic language-based interactive decision-making tasks in household environments.

With AutoGen, we implemented a two-agent system to solve tasks from ALFWorld. It consists of an LLM-backed assistant agent responsible for suggesting plans to conduct a task and an executor agent responsible for executing actions in the ALFWorld environments. This system integrates ReAct prompting (Yao et al., 2022), and is able to achieve similar performance. A common challenge encountered in both ReAct and the AutoGen-based two-agent system is their occasional inability to leverage basic commonsense knowledge about the physical world. This deficiency can lead to the system getting stuck in a loop due to repetitive errors. Fortunately, the modular design of AutoGen allows us to address this issue effectively: With AutoGen, we are able to introduce a grounding agent, which supplies crucial commonsense knowledge—such as “*You must find and take the object before you can examine it. You must go to where the target object is before you can use it.*”—whenever the system exhibits early signs of recurring errors. It significantly enhances the system’s ability to avoid getting entangled in error loops. We compare the task-solving performance of the two variants of our system with GPT-3.5-turbo and ReAct⁷ on the 134 unseen tasks from ALFWorld and report results in Figure 4c. The results show that introducing a grounding agent could bring in a 15% performance gain on average. Upon examining the systems’ outputs, we observe that the grounding agent, by delivering background commonsense knowledge at the right junctures, significantly mitigated the tendency of the system to persist with a flawed plan, thereby avoiding the creation of error loops. For an example trajectory comparing the systems see Appendix D, Figure 10.

A4: Multi-Agent Coding

In this subsection, we use AutoGen to build a multi-agent coding system based on OptiGuide (Li et al., 2023a), a system that excels at writing code to interpret optimization solutions and answer user questions, such as exploring the implications of changing a supply-chain decision or understanding why the optimizer made a particular choice. The second sub-figure of Figure 3 shows the AutoGen-based implementation. The workflow is as follows: the end user sends questions, such as “*What if we prohibit shipping from supplier 1 to roastery 2?*” to the Commander agent. The Commander coordinates with two assistant agents, including the Writer and the Safeguard, to answer the question. The Writer will craft code and send the code to the Commander. After receiving the code, the Commander checks the code safety with the Safeguard; if cleared, the Commander will use external tools (e.g., Python) to execute the code, and request the Writer to interpret the execution results. For instance, the writer may say “*if we prohibit shipping from supplier 1 to roastery 2, the total cost would increase by 10.5%.*” The Commander then provides this concluding answer to the end user. If, at a particular step, there is an exception, e.g., security red flag raised by Safeguard, the Commander redirects the issue back to the Writer with debugging information. The process might be repeated multiple times until the user’s question is answered or timed-out.

With AutoGen the core workflow code for OptiGuide was reduced from over 430 lines to 100 lines, leading to significant productivity improvement. We provide a detailed comparison of user experience with ChatGPT+Code Interpreter and AutoGen-based OptiGuide in Appendix D, where we show that AutoGen-based OptiGuide could save around 3x of user’s time and reduce user interactions by 3 - 5 times on average. We also conduct an ablation showing that multi-agent abstraction is necessary. Specifically, we construct a single-agent approach where a single agent conducts both the code-writing and safeguard processes. We tested the single- and multi-agent approaches on a dataset of 100 coding tasks, which is crafted to include equal numbers of safe and unsafe tasks. Evaluation results as reported in Figure 4d show that the multi-agent design boosts the F-1 score in identifying unsafe code by 8% (with GPT-4) and 35% (with GPT-3.5-turbo).

⁷Results of ReAct are obtained by directly running its official code with default settings. The code uses `text-davinci-003` as backend LM and does not support GPT-3.5-turbo or GPT-4.

A5: Dynamic Group Chat

AutoGen provides native support for a *dynamic group chat* communication pattern, in which participating agents share the same context and converse with the others in a dynamic manner instead of following a pre-defined order. Dynamic group chat relies on ongoing conversations to guide the flow of interaction among agents. These make dynamic group chat ideal for situations where collaboration without strict communication order is beneficial. In AutoGen, the `GroupChatManager` class serves as the conductor of conversation among agents and repeats the following three steps: dynamically selecting a speaker, collecting responses from the selected speaker, and broadcasting the message (Figure 3-A5). For the dynamic speaker-selection component, we use a role-play style prompt. Through a pilot study on 12 manually crafted complex tasks, we observed that compared to a prompt that is purely based on the task, utilizing a role-play prompt often leads to more effective consideration of both conversation context and role alignment during the problem-solving and speaker-selection process. Consequently, this leads to a higher success rate and fewer LLM calls. We include detailed results in Appendix D.

A6: Conversational Chess

Using AutoGen, we developed Conversational Chess, a natural language interface game shown in the last sub-figure of Figure 3. It features built-in agents for players, which can be human or LLM, and a third-party board agent to provide information and validate moves based on standard rules.

With AutoGen, we enabled two essential features: (1) Natural, flexible, and engaging game dynamics, enabled by the customizable agent design in AutoGen. Conversational Chess supports a range of game-play patterns, including AI-AI, AI-human, and human-human, with seamless switching between these modes during a single game. An illustrative example of these entertaining game dynamics can be found in Figure 15, Appendix D. (2) Grounding, which is a crucial aspect to maintain game integrity. During gameplay, the board agent checks each proposed move for legality; if a move is invalid, the agent responds with an error, prompting the player agent to re-propose a legal move before continuing. This process ensures that only valid moves are played and helps maintain a consistent gaming experience. As an ablation study, we removed the board agent and instead only relied on a relevant prompt “*you should make sure both you and the opponent are making legal moves*” to ground their move. The results highlighted that without the board agent, illegitimate moves caused game disruptions. The modular design offered flexibility, allowing swift adjustments to the board agent in response to evolving game rules or varying chess rule variants. A comprehensive demonstration of this ablation study is in Appendix D.

4 Discussion

We introduced an open-source library, AutoGen, that incorporates the paradigms of conversable agents and conversation programming. This library utilizes capable agents that are well-suited for multi-agent cooperation. It features a unified conversation interface among the agents, along with an auto-reply mechanisms, which help establish an agent-interaction interface that capitalizes on the strengths of chat-optimized LLMs with broad capabilities while accommodating a wide range of applications. AutoGen serves as a general framework for creating and experimenting with multi-agent systems that can easily fulfill various practical requirements, such as reusing, customizing, and extending existing agents, as well as programming conversations between them.

Our experiments, as detailed in Section 3, demonstrate that this approach offers numerous benefits. The adoption of AutoGen has resulted in improved performance (over state-of-the-art approaches), reduced development code, and decreased manual burden for existing applications. It offers flexibility to developers, as demonstrated in A1 (scenario 3), A5, and A6, where AutoGen enables multi-agent chats to follow a dynamic pattern rather than fixed back-and-forth interactions. It allows humans to engage in activities alongside multiple AI agents in a conversational manner. Despite the complexity of these applications (most involving more than two agents or dynamic multi-turn agent cooperation), the implementation based on AutoGen remains straightforward. Dividing tasks among separate agents promotes modularity. Furthermore, since each agent can be developed, tested, and maintained separately, this approach simplifies overall development and code management.

Although this work is still in its early experimental stages, it paves the way for numerous future directions and research opportunities. For instance, we can explore effective integration of existing agent implementations into our multi-agent framework and investigate the optimal balance between automation and human control in multi-agent workflows. As we further develop and refine AutoGen, we aim to investigate which strategies, such as agent topology and conversation patterns, lead to the most effective multi-agent conversations while optimizing the overall efficiency, among other factors. While increasing the number of agents and other degrees of freedom presents opportunities for tackling more complex problems, it may also introduce new safety challenges that require additional studies and careful consideration.

We provide more discussion in Appendix B, including guidelines for using AutoGen and direction of future work. We hope AutoGen will help improve many LLM applications in terms of speed of development, ease of experimentation, and overall effectiveness and safety. We actively welcome contributions from the broader community.

Ethics statement

There are several potential ethical considerations that could arise from the development and use of the AutoGen framework.

- **Privacy and Data Protection:** The framework allows for human participation in conversations between agents. It is important to ensure that user data and conversations are protected, and that developers use appropriate measures to safeguard privacy.
- **Bias and Fairness:** LLMs have been shown to exhibit biases present in their training data (Navigli et al., 2023). When using LLMs in the AutoGen framework, it is crucial to address and mitigate any biases that may arise in the conversations between agents. Developers should be aware of potential biases and take steps to ensure fairness and inclusivity.
- **Accountability and Transparency:** As discussed in the future work section, as the framework involves multiple agents conversing and cooperating, it is important to establish clear accountability and transparency mechanisms. Users should be able to understand and trace the decision-making process of the agents involved in order to ensure accountability and address any potential issues or biases.
- **Trust and Reliance:** AutoGen leverages human understanding and intelligence while providing automation through conversations between agents. It is important to consider the impact of this interaction on user experience, trust, and reliance on AI systems. Clear communication and user education about the capabilities and limitations of the system will be essential (Cai et al., 2019).
- **Unintended Consequences:** As discussed before, the use of multi-agent conversations and automation in complex tasks may have unintended consequences. In particular, allowing LLM agents to make changes in external environments through code execution or function calls, such as installing packages, could be risky. Developers should carefully consider the potential risks and ensure that appropriate safeguards are in place to prevent harm or negative outcomes.

Acknowledgements

The work presented in this report was made possible through discussions and feedback from Peter Lee, Johannes Gehrke, Eric Horvitz, Steven Lucco, Umesh Madan, Robin Moeur, Piali Choudhury, Saleema Amershi, Adam Fournay, Victor Dibia, Guoqing Zheng, Corby Rosset, Ricky Loynd, Ece Kamar, Rafah Hosn, John Langford, Ida Momennejad, Brian Krabach, Taylor Webb, Shanka Subhra Mondal, Wei-ge Chen, Robert Gruen, Yinan Li, Yue Wang, Suman Nath, Tanakorn Leesatapornwongsa, Xin Wang, Shishir Patil, Tianjun Zhang, Saehan Jo, Ishai Menache, Kontantina Meliou, Runlong Zhou, Feiran Jia, Hamed Khanpour, Hamid Palangi, Srinagesh Sharma, Julio Albinati Cortez, Amin Saied, Yuzhe Ma, Dujian Ding, Linyong Nan, Prateek Yadav, Shannon Shen, Ankur Mallick, Mark Encarnación, Lars Liden, Tianwei Yue, Julia Kiseleva, Anastasia Razdaibiedina, and Luciano Del Corro. Qingyun Wu would like to acknowledge the funding and research support from the College of Information Science and Technology at Penn State University.